

Digital Signal Processing

Project Report

Speech Equalizer & ECG Signal Processing

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Abstract

This report presents two DSP applications developed as part of the Digital Signal Processing course project. The first application implements a real-time speech equalizer for audio signal enhancement using multi-band FIR filters designed with the Hamming window method. The second application performs ECG signal denoising using a combination of a 50 Hz notch filter and a low-pass FIR filter to suppress power-line interference and high-frequency muscle noise.

Both systems were implemented and validated using MATLAB. Filter characteristics, frequency responses, time-domain comparisons, and spectrograms are presented and analyzed throughout this report.

Part 1: Speech Equalizer — Mohamed Salah

The speech equalizer processes an audio signal recorded at a 22,050 Hz sampling rate. A three-band FIR equalizer was designed using the Hamming window method, allowing independent gain control over low (0–100 Hz), mid (100–300 Hz), and high (300–800 Hz) frequency bands.

1.1 Sample Rate Conversion

The input audio signal was analyzed at three different sampling rates: original (22,050 Hz), upsampled 4× (88,200 Hz), and downsampled to half-rate (11,025 Hz). This demonstrates the impact of resampling on signal quality and filter behavior.

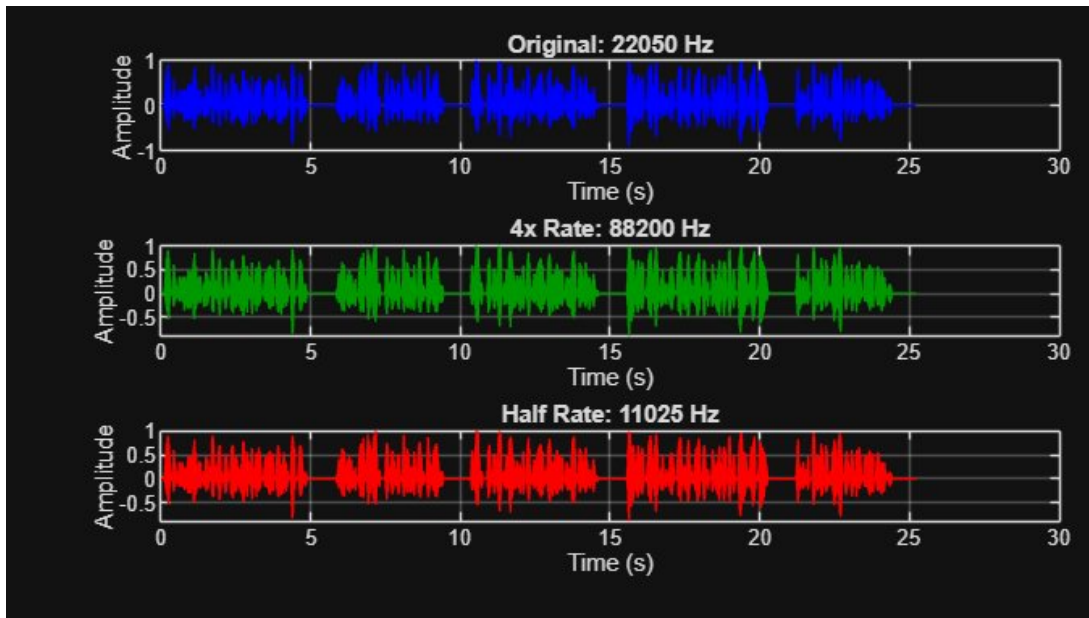


Figure 1.1 — Time-domain comparison: original (22,050 Hz), 4× upsampled (88,200 Hz), and half-rate (11,025 Hz) signals

1.2 Multi-Band FIR Filter Design

Three order-100 FIR band-pass filters were designed using the Hamming window to cover each equalization band. All filters exhibit linear phase response — a key advantage of FIR design that ensures zero phase distortion.

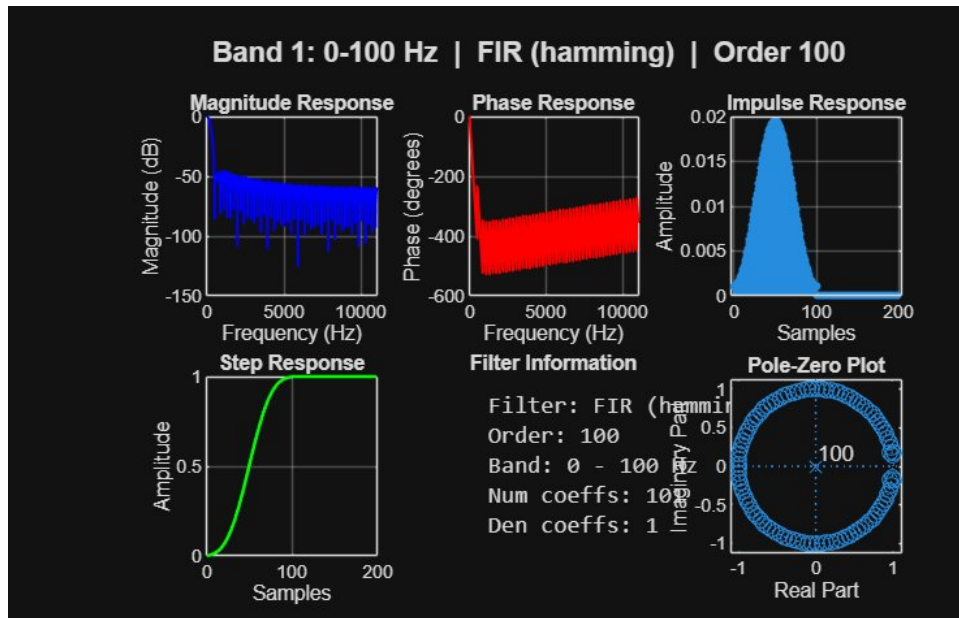


Figure 1.2 — Band 1 (0–100 Hz): Magnitude, phase, impulse & step response, pole-zero plot

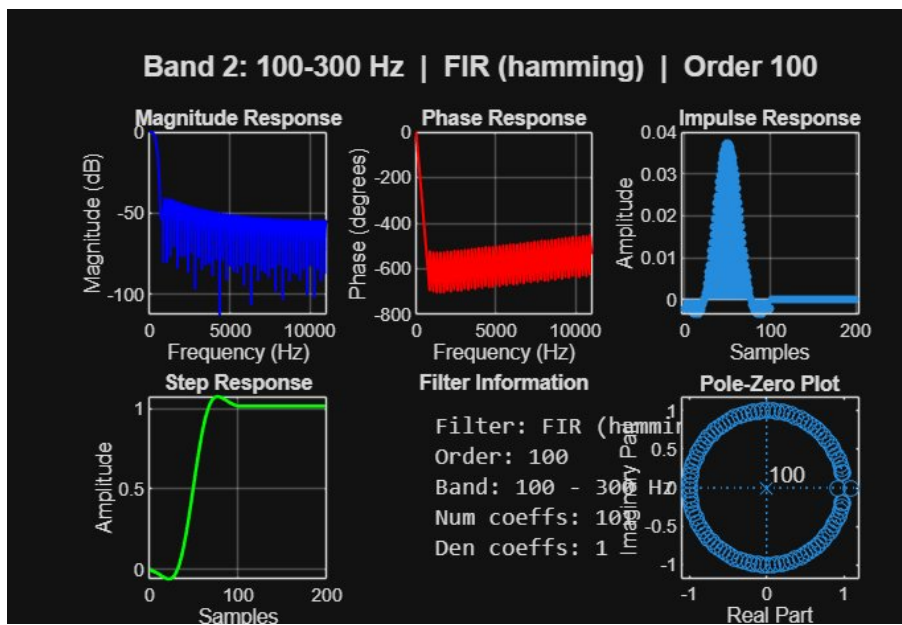


Figure 1.3 — Band 2 (100–300 Hz): Magnitude, phase, impulse & step response, pole-zero plot

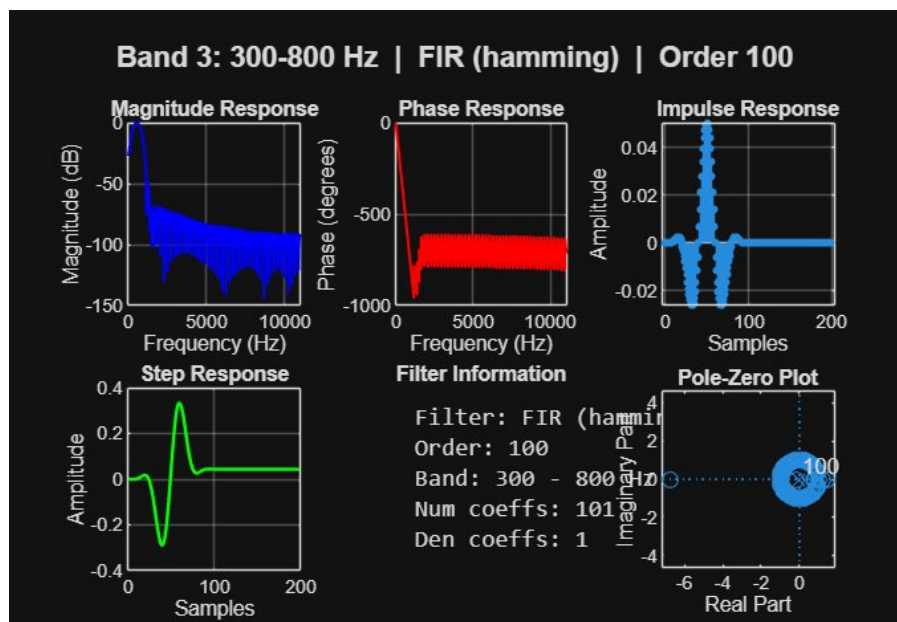


Figure 1.4 — Band 3 (300–800 Hz): Magnitude, phase, impulse & step response, pole-zero plot

1.3 Time & Frequency Domain Analysis

After applying the equalizer pipeline, the processed signal was compared against the original in both time and frequency domains. The power spectral density confirms that the equalizer redistributes spectral energy according to the configured band gains. The spectrogram provides a time-frequency visualization revealing how harmonic content is preserved and reshaped.

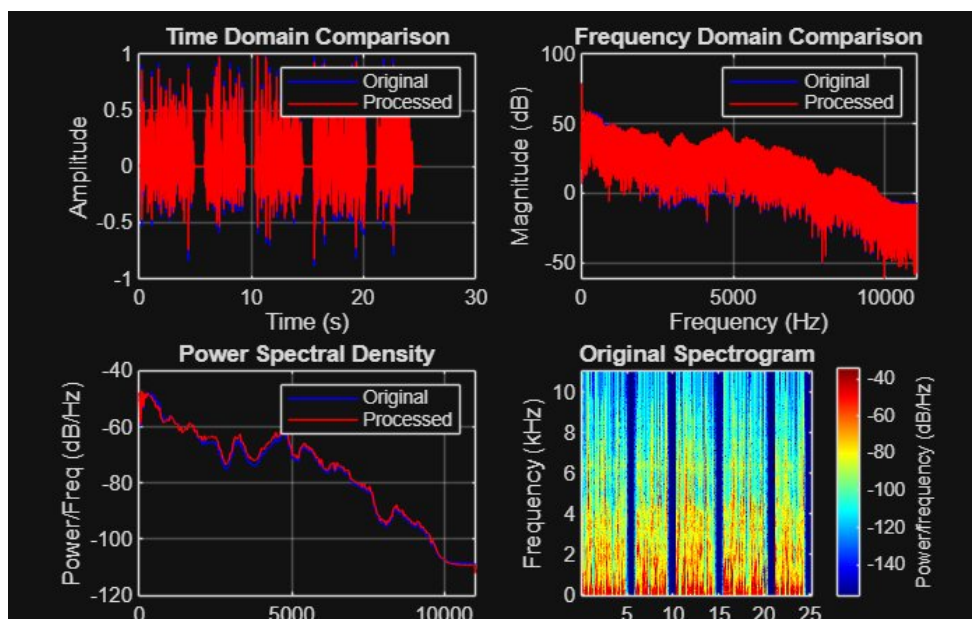


Figure 1.5 — Time-domain, frequency-domain, PSD comparison, and original spectrogram

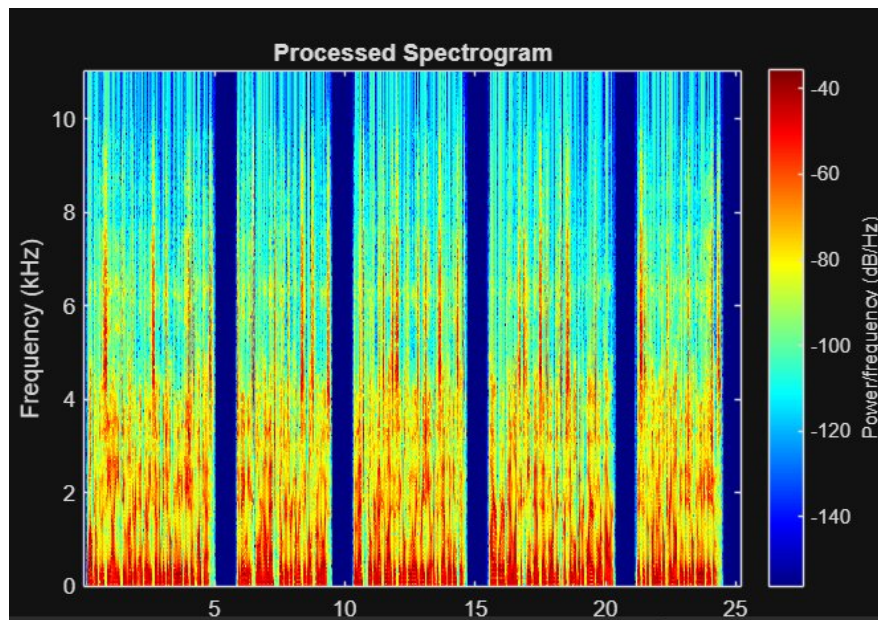


Figure 1.6 — Processed spectrogram showing energy distribution after equalization

Part 2: ECG Signal Processing — Mohamed Hossam

The ECG signal processing component targets two common noise sources in clinical recordings: 50 Hz power-line interference and high-frequency electromyographic (EMG) muscle noise. Two PhysioNet MIT-BIH Arrhythmia Database records were used — Record 100 (normal sinus rhythm) and Record 106 (abnormal rhythm with ventricular ectopy) — to validate the denoising pipeline on both signal types.

2.1 Filter Design

A second-order IIR notch filter was designed at 50 Hz to eliminate power-line interference with minimal distortion to adjacent frequencies. A linear-phase FIR low-pass filter was then applied to remove high-frequency muscle noise, with a cutoff at 100 Hz, appropriate for ECG diagnostics.

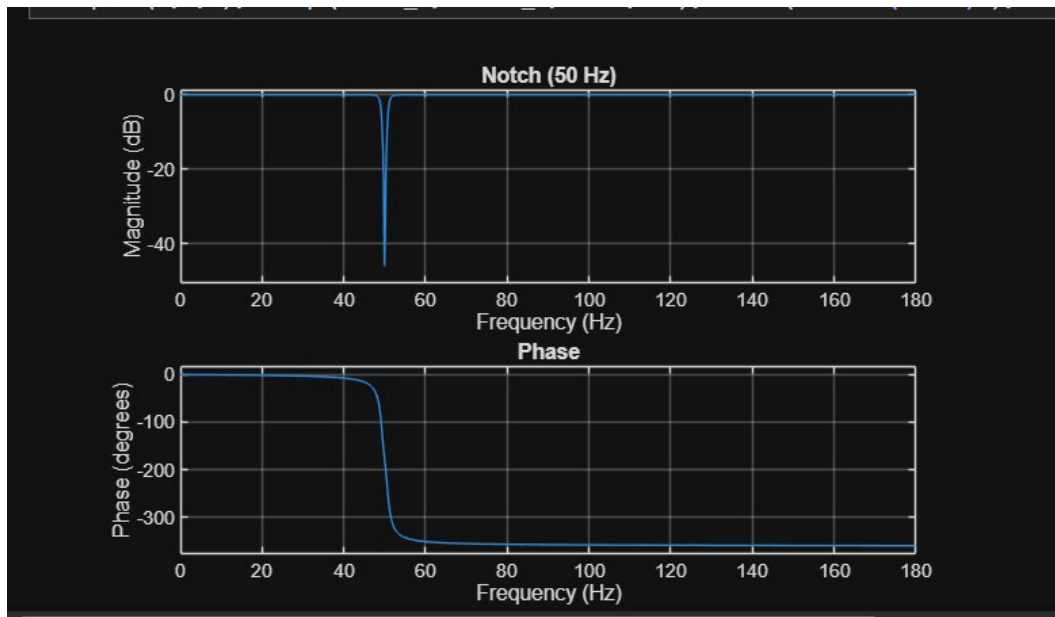


Figure 2.1 — Notch filter (50 Hz): Magnitude and phase response

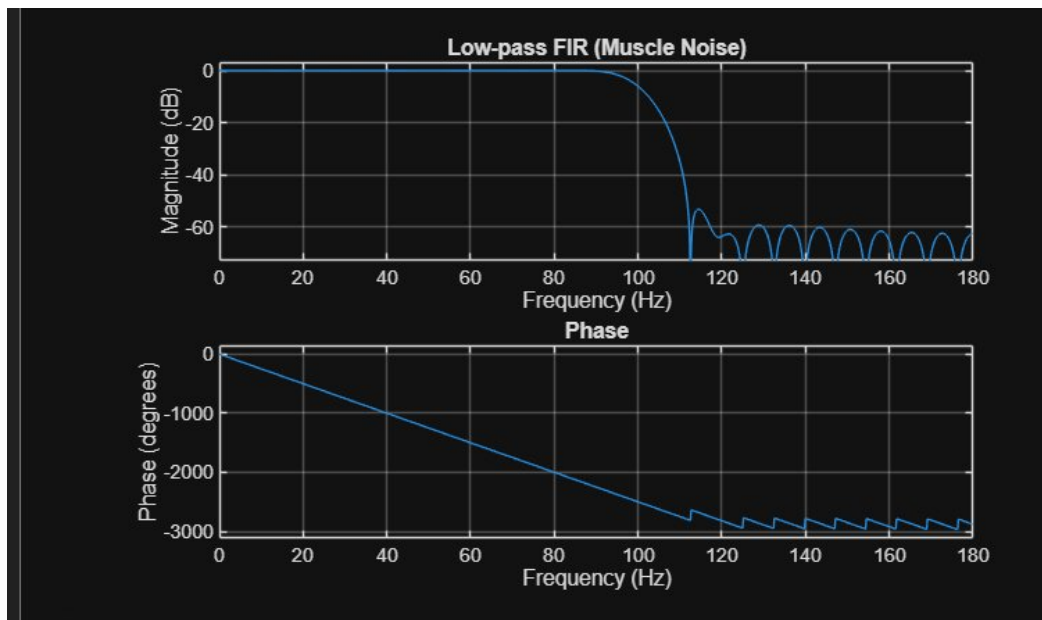


Figure 2.2 — Low-pass FIR filter for muscle noise removal: Magnitude and phase response

2.2 ECG Denoising Results

The fully denoised ECG signals are compared against their original noisy counterparts in the time domain. Record 100 shows clearly defined QRS complexes and P/T waves after denoising. Record 106 exhibits the expected irregular ventricular rhythm with high-amplitude ectopic beats, which remain clearly visible after noise suppression.

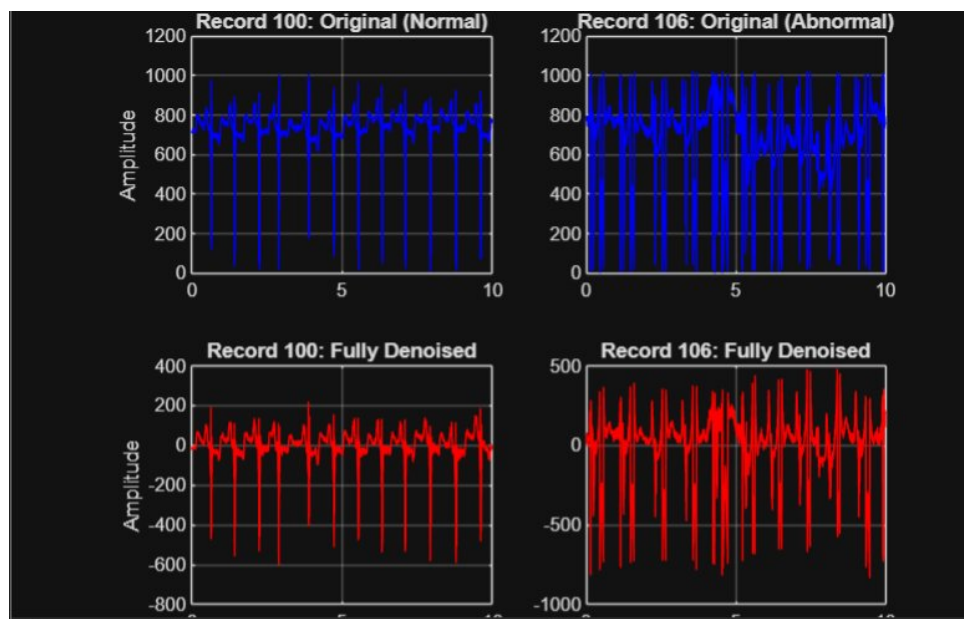


Figure 2.3 — Original (blue) and fully denoised (red) ECG signals for Records 100 and 106

2.3 Power Spectral Density Analysis

The PSD comparison confirms a sharp attenuation at 50 Hz while preserving the diagnostic frequency band (0–40 Hz). The low-pass filter removes residual high-frequency noise above 100 Hz. In Record

106, the abnormal signal energy distribution highlights the importance of broadband denoising for arrhythmia detection.

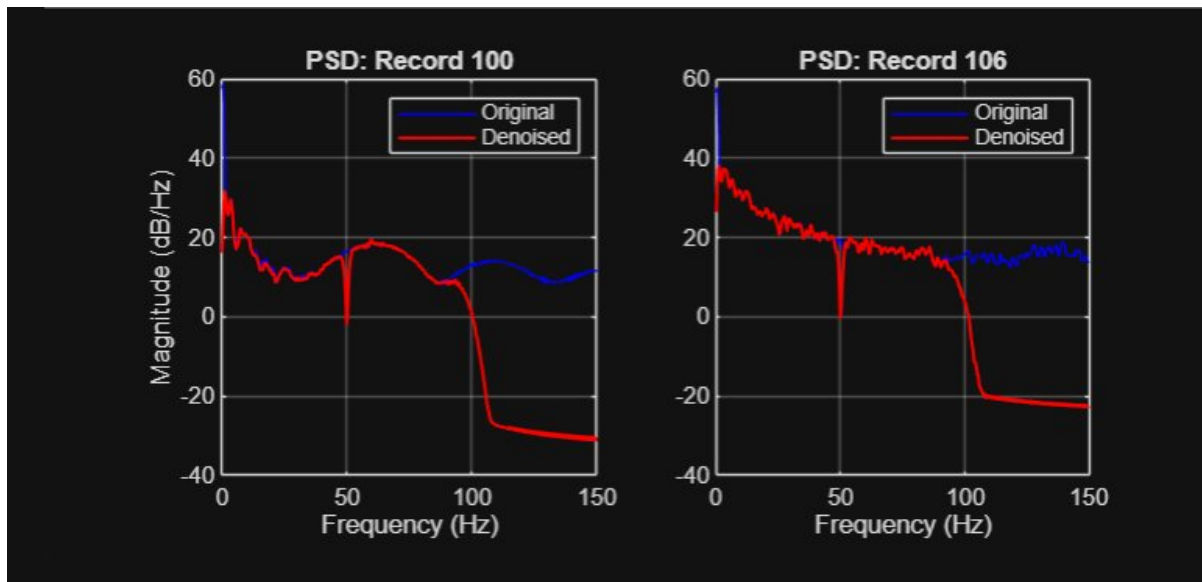


Figure 2.4 — PSD comparison (original vs. denoised) for Records 100 and 106

Conclusion

This project successfully implemented and evaluated two DSP applications:

Speech Equalizer (Mohamed Salah): A three-band FIR equalizer using Hamming-windowed filters was designed and tested. The multi-rate analysis demonstrated the relationship between sampling rate and signal bandwidth. Spectrogram and PSD analyses confirmed effective band-selective gain control.

ECG Denoiser (Mohamed Hossam): A cascade of an IIR notch filter at 50 Hz and an FIR low-pass filter effectively suppressed power-line interference and muscle noise from both normal and abnormal ECG recordings. The denoised signals clearly preserve clinically relevant waveform morphology.

Both systems demonstrate the practical power of digital filter design in real-world biomedical and audio processing applications. MATLAB proved an effective platform for rapid prototyping and visualization of DSP algorithms.

References

- [1] Proakis, J. G., & Manolakis, D. G. (2006). *Digital Signal Processing: Principles, Algorithms, and Applications* (4th ed.). Prentice Hall.
- [2] PhysioNet. MIT-BIH Arrhythmia Database. <https://physionet.org/content/mitdb/>
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- [4] Oppenheim, A. V., & Schaffer, R. W. (2009). *Discrete-Time Signal Processing* (3rd ed.). Prentice Hall.